INCH-POUND

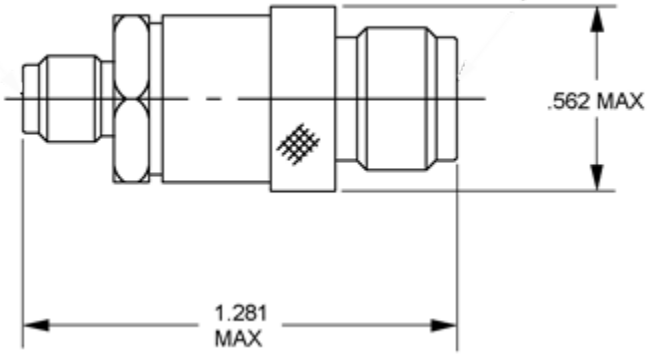
MIL-PRF-55339/41B
w/AMENDMENT 1
16 November 2023
SUPERSEDING
MIL-PRF-55339/41B
28 January 2015

PERFORMANCE SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA (FEMALE) TO SERIES TNC (FEMALE)), CLASS 2 STRAIGHT PLUG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55339.



NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial representations are for reference purposes only.
3. Interface shall be in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

FSC 5935

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ENGINEERING DATA:

Nominal impedance: 50 ohms.
Frequency range: 0 to 11 GHz.
Voltage rating: 335 volts rms maximum working voltage at sea level, 85 volts rms at 70,000 feet.
Temperature range: -65° to +165°C.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Center contact retention.
Axial force – 6 pounds minimum.

Force to engage and disengage.
Longitudinal force – Not applicable.
Torque – 2 inch-pounds maximum.

Coupling proof torque.
Series SMA – Not applicable.
Series TNC – Not applicable.

Inspection conditions.
Coupling torque – Series SMA – 7 to 10 inch-pounds.
Series TNC – 4 to 6 inch-pounds.

Mating characteristics.
Center contact (socket):
Oversize test pin. Series SMA - .0375 +.0001, Series TNC .057 diameter minimum (nonenclosed entry contacts only).
Insertion depth – Series SMA .030/.045, Series TNC .125 minimum.
Number of insertions – Series SMA 3, Series TNC 1.

Insertion force test. Steel test pin diameter Series SMA .0370+.0001. Series TNC - .054 inch minimum.
Test pin finish – 16 microinches.
Insertion force – Series SMA 3 pounds maximum. Series TNC 2 pounds maximum.

Withdrawal force. Steel test pin diameter Series SMA .0355 - .0001, Series TNC - .052 inch maximum.
Withdrawal force – Series SMA 1 ounce minimum, Series TNC 2 ounces minimum.
Test pin finish – 16 microinches.

Permeability: Not to exceed 2.

Insulation resistance. 5,000 megohms minimum.

Voltage standing wave ratio (VSWR): 1.15 + .015 F (GHz) at .5 to 11 GHz.

RF leakage: -60 dB minimum tested at a frequency of 2 to 3 GHz.

RF insertion loss: $.06\sqrt{F\ (GHz)}$ dB maximum at 6 GHz.

Durability: 500 cycles minimum at 12 cycles per minute maximum.

Dielectric withstanding voltage: 1,000 volts rms minimum at sea level.

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Contact resistance (milliohms, max):

	Initial	After environment
Center contact	4.5 1/	6.0
Outer contact	2.2	--

1/ Two center contacts must be mated to the center conductor under test therefore doubling center contact resistance.

Vibration, high frequency: MIL-STD-202-204, test condition D, interruptions – 1 µs, maximum.

Shock (specified pulse): MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition C.

Moisture resistance: 200 megohms, minimum.

Corona level:
Voltage – 375 volts minimum.
Altitude – 70,000 feet minimum.

RF high potential withstanding voltage:
RF voltage – 1,000 volts rms minimum.
Frequency – 5 to 7.5 MHz.

Salt spray (corrosion): MIL-STD-202-101, test condition B.

Part or Identifying Number (PIN): M55339/41-30001
M55339/41-50001

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-DTL-55339, this document references the following:

MIL-STD-202-101
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-348

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2024-003)

Review activities:
Army - AR, AT, EA
Navy - AS, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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